THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Assertion (A): Rusting of iron is a chemical change. Reason (R): A new substance with different properties (iron oxide) is formed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Dry HCl gas does not change the colour of dry blue litmus paper. Reason (R): HCl gas does not ionise in the absence of water to release H^+ ions. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Sodium is stored in kerosene oil. Reason (R): Sodium reacts vigorously with oxygen and moisture of the air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): The rate of breathing in aquatic organisms is much faster than in terrestrial organisms. Reason (R): The amount of dissolved oxygen in water is fairly low compared to air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): Reflex actions are not controlled by the brain directly. Reason (R): The reflex arc connects the receptor to the effector through the spinal cord. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Asexual reproduction produces offspring that are not exactly identical to the parent. Reason (R): Copying of DNA is accompanied by small errors. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): Sex of a child is determined by the father. Reason (R): Sperms carry either X or Y chromosome. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A convex mirror is used as a rear-view mirror. Reason (R): It always forms virtual, erect and diminished images with a wide field of view. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The image formed on the retina is inverted. Reason (R): The brain interprets the signals to make the object appear erect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): A voltmeter is connected in parallel with a resistor. Reason (R): A voltmeter has a very high resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Magnetic field lines never intersect. Reason (R): The field at a point has only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): The number of trophic levels in a food chain is limited. Reason (R): About 90 per cent of energy is lost at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 28 g **(b)** 20 g **(c)** 12 g **(d)** 44 g

15. Equal volumes of HCl and NaOH solutions of the same concentration are mixed. The resulting solution has pH: 1

(a) 13 **(b)** 7 **(c)** 3 **(d)** 1

16. Metal X reacts with dilute HCl giving H_2 but does not react with cold or hot water. Metal Y reacts with cold water. Which is more reactive? 1

(a) cannot be said **(b)** Y **(c)** both equally **(d)** X

17. The balanced equation for the complete combustion of methane is: 1

(a) $CH_4 + O_2 \longrightarrow CO_2 + H_2O$ **(b)** $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(c)** $2\,CH_4 + O_2 \longrightarrow 2\,CO_2 + H_2O$ **(d)** $CH_4 + 3\,O_2 \longrightarrow CO_2 + 2\,H_2O$

18. In aerobic respiration, one molecule of glucose gives how many molecules of carbon dioxide? 1

(a) 12 **(b)** 2 **(c)** 6 **(d)** 1

19. A hormone X is secreted by an organ below the stomach, regulates sugar. X's deficiency causes: 1

(a) gigantism **(b)** goitre **(c)** dwarfism **(d)** diabetes

20. A plant breeder crosses two varieties and also grows a plant by stem cutting. The offspring from the cross show 1

(a) identical traits in the cross
(b) no variation in either
(c) variation, and those from the cutting are identical to the parent
(d) variation in the cutting only

21. A pea plant RrYy is crossed with rryy. Out of 200 offspring, the number of round green plants is expected to be 1

(a) 25 **(b)** 100 **(c)** 50 **(d)** 150

22. A concave mirror forms a real image of a candle on a screen kept at 45 cm. If the candle is 15 cm from the mirror, the focal length of the mirror (magnitude) is 1

(a) 30 cm **(b)** 15 cm **(c)** 22.5 cm **(d)** 11.25 cm

23. A person with a far point at 25 cm requires spectacles to see distant objects clearly. The focal length of the lens is 1

(a) -6 m **(b)** +25 cm **(c)** -25 cm **(d)** -31 cm

24. The resistance of a wire of length L and area A is 12 ohm. A second wire of the same material has length 2L and radius double that of the first. Its resistance is 1

(a) 6 ohm **(b)** 3 ohm **(c)** 24 ohm **(d)** 12 ohm

25. A straight wire with current is placed between the poles of a magnet. The force on the wire is greatest when the wire makes an angle with the field of 1

(a) 30 degrees **(b)** 0 degrees **(c)** 45 degrees **(d)** 90 degrees

26. A food chain has four trophic levels. If the energy at the fourth level is 5 J, the energy at the first level is 1
(a) 50 J **(b)** 50,000 J **(c)** 500 J **(d)** 5000 J

27. A metal X turns black on heating in air. The black product, on passing hydrogen over it, turns reddish brown. X is: 1
(a) zinc **(b)** magnesium **(c)** copper **(d)** iron

28. A white powder X on adding water forms a hard solid mass. X is: 1
(a) bleaching powder **(b)** baking soda **(c)** washing soda **(d)** plaster of Paris

29. A student heats a green carbonate ore of copper. The black residue and gas are: 1
(a) CuO and CO_2 **(b)** Cu_2O and O_2
(c) CuO and CO **(d)** Cu and CO_2

30. A hydrocarbon X decolourises bromine water and has a molecular mass of 28 u. X is: 1
(a) ethane **(b)** ethene **(c)** ethyne **(d)** methane

31. A closed jar contains a green plant in sunlight together with a mouse. The mouse survives longer than a mouse alone in a jar because the plant: 1
(a) absorbs oxygen only **(b)** releases carbon dioxide
(c) uses carbon dioxide and releases oxygen **(d)** gives out nitrogen

32. A boy has a rapid pulse, sweating and wide pupils when frightened. This is due to: 1
(a) adrenaline **(b)** thyroxine **(c)** oestrogen **(d)** insulin

33. An organism with a very complex body that can still regenerate would need 1
(a) only gametes
(b) undifferentiated cells to produce missing parts
(c) no DNA
(d) no cells at all

34. A pea plant that is tall and round (TtRr) is crossed with dwarf wrinkled (ttrr). The offspring ratio is 1
(a) 9 : 3 : 3 : 1 **(b)** 1 : 1 **(c)** 3 : 1 **(d)** 1 : 1 : 1 : 1

35. An object is placed 24 cm from a lens and a real image forms at 24 cm. The lens is 1
(a) convex, f = 12 cm **(b)** concave, f = -12 cm
(c) convex, f = 24 cm **(d)** concave, f = -24 cm

36. The sky is blue because 1
(a) the sea reflects blue
(b) smaller wavelengths are scattered more by small particles
(c) the sun emits blue only
(d) larger wavelengths are scattered more by small particles

37. Which graph is a straight line through the origin for a metallic conductor at constant temperature? 1
(a) V against t **(b)** V against I **(c)** I against R **(d)** P against I

38. A generator coil of frequency 50 Hz is rotated at double the speed. The frequency becomes 1
(a) 25 Hz **(b)** 100 Hz **(c)** 200 Hz **(d)** 50 Hz

39. Ozone in the stratosphere is formed by the action of 1
(a) CFCs on oxygen **(b)** infrared on ozone
(c) UV radiation on oxygen **(d)** visible light on nitrogen

40. Assertion (A): Decomposition of calcium carbonate is an endothermic reaction. Reason (R): Heat energy is absorbed to break the bonds in the compound. 1
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. **Assertion (A):** Burning of magnesium ribbon in air is a combination reaction. 1
Reason (R): In a combination reaction two or more substances combine to form a single product.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Dry hydrogen chloride gas does not change the colour of dry blue litmus paper. 1
Reason (R): Acids produce H^+ ions only in the presence of water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Sodium is stored under kerosene oil. 1
Reason (R): Sodium reacts vigorously with oxygen and moisture of the air and may catch fire.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** Carbon forms a very large number of compounds. 1
Reason (R): Carbon is tetravalent and shows the property of catenation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Guard cells control the opening and closing of stomata. 1
Reason (R): Guard cells swell when water flows into them, causing the stomatal pore to open.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** A reflex action is a quick, automatic response. 1
Reason (R): In a reflex action the signal passes through the spinal cord and not through the brain.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Plants raised by vegetative propagation show characters identical to the parent plant. 1
Reason (R): Vegetative propagation involves only one parent and no fusion of gametes.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** When pure tall (TT) and pure dwarf (tt) pea plants are crossed, all the F1 plants are tall. 1
Reason (R): Tallness is the dominant trait over dwarfness.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A convex mirror is used as a rear-view mirror in vehicles. 1
Reason (R): A convex mirror always forms an erect, diminished image and gives a wider field of view.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Myopia is corrected by using a concave lens. 1
Reason (R): A concave lens diverges the rays so that the image is formed farther back, on the retina.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**